

6.1.11 Combined bending moment and local load or support reaction

(1) Cross-sections subject to the combined action of a bending moment M_{Ed} and a transverse force due to a local load or support reaction F_{Ed} should satisfy the following:

$$M_{Ed} / M_{c,Rd} \leq 1 \quad \dots (6.28a)$$

$$F_{Ed} / R_{w,Rd} \leq 1 \quad \dots (6.28b)$$

$$\frac{M_{Ed}}{M_{c,Rd}} + \frac{F_{Ed}}{R_{w,Rd}} \leq 1,25 \quad \dots (6.28c)$$

where:

$M_{c,Rd}$ is the moment resistance of the cross-section given in 6.1.4.1(1);

$R_{w,Rd}$ is the appropriate value of the local transverse resistance of the web from 6.1.7.

In equation (6.28c) the bending moment M_{Ed} may be calculated at the edge of the support. For members and sheeting with more than one web, $R_{w,Rd}$ is the sum of the local transverse resistances of the individual webs.

6.2 Buckling resistance

6.2.1 General

- (1) In members with cross-sections that are susceptible to cross-sectional distortion, account should be taken of possible lateral buckling of compression flanges and lateral bending of flanges generally.
- (2) The effects of local and distortional buckling should be taken into account as specified in Section 5.5.

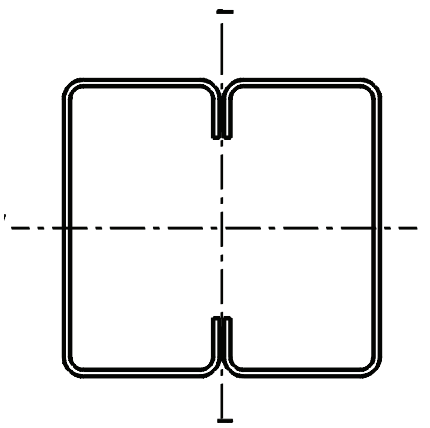
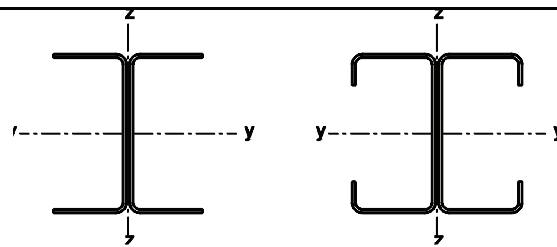
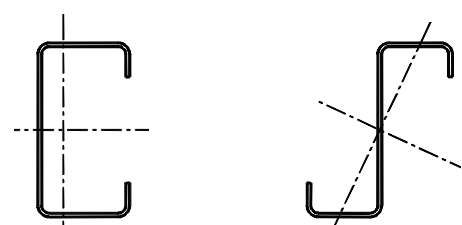
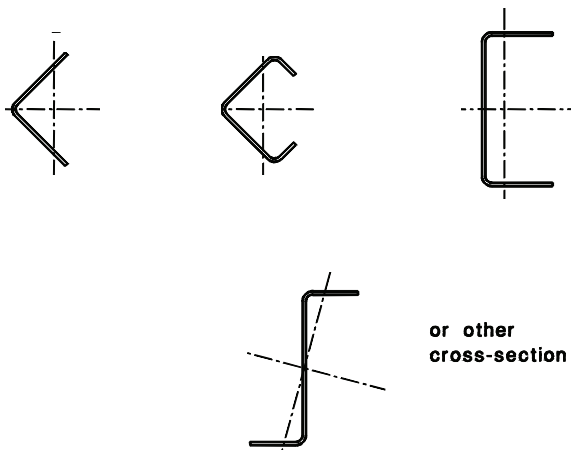
6.2.2 Flexural buckling

- (1) The design buckling resistance $N_{b,Rd}$ for flexural buckling should be obtained from EN 1993-1-1 using the appropriate buckling curve from table 6.3 according to the type of cross-section, axis of buckling and yield strength used, see (3).
- (2) The buckling curve for a cross-section not included in table 6.3 may be obtained by analogy.
- (3) The buckling resistance of a closed built-up cross-section should be determined using either:
 - buckling curve b in association with the basic yield strength f_{yb} of the flat sheet material out of which the member is made by cold forming;
 - buckling curve c in association with the average yield strength f_{ya} of the member after cold forming, determined as specified in 3.2.3, provided that $A_{eff} = A_g$.

6.2.3 Torsional buckling and torsional-flexural buckling

- (1) For members with point-symmetric open cross-sections (e.g Z-purlin with equal flanges), account should be taken of the possibility that the resistance of the member to torsional buckling might be less than its resistance to flexural buckling.
- (2) For members with mono-symmetric open cross-sections, see figure 6.12, account should be taken of the possibility that the resistance of the member to torsional-flexural buckling might be less than its resistance to flexural buckling.
- (3) For members with non-symmetric open cross-sections, account should be taken of the possibility that the resistance of the member to either torsional or torsional-flexural buckling might be less than its resistance to flexural buckling.
- (4) The design buckling resistance $N_{b,Rd}$ for torsional or torsional-flexural buckling should be obtained from EN 1993-1-1, 6.3.1.1 using the relevant buckling curve for buckling about the z-z axis obtained from table 6.3.

Table 6.3: Appropriate buckling curve for various types of cross-section

Type of cross-section		Buckling about axis	Buckling curve
	if f_{yb} is used	Any	b
	if f_{ya} is used ^{*)}	Any	c
		y - y	a
		z - z	b
		Any	b
 or other cross-section		Any	c
^{*)} The average yield strength f_{ya} should not be used unless $A_{eff} = A_g$			

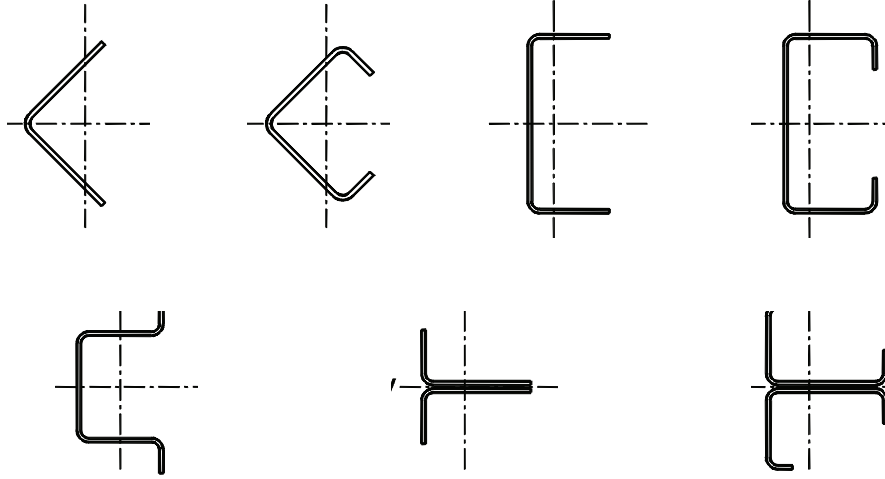


Figure 6.12: Monosymmetric cross-sections susceptible to torsional-flexural buckling

(5) The elastic critical force $N_{cr,T}$ for torsional buckling of simply supported beam should be determined from:

$$N_{cr,T} = \frac{1}{i_o^2} \left(G I_t + \frac{\pi^2 E I_w}{l_T^2} \right) \quad \dots (6.33a)$$

with:

$$i_o^2 = i_y^2 + i_z^2 + y_o^2 + z_o^2 \quad \dots (6.33b)$$

where:

G is the shear modulus;

I_t is the torsion constant of the gross cross-section;

I_w is the warping constant of the gross cross-section;

i_y is the radius of gyration of the gross cross-section about the y - y axis;

i_z is the radius of gyration of the gross cross-section about the z - z axis;

l_T is the buckling length of the member for torsional buckling;

y_o, z_o are the shear centre co-ordinates with respect to the centroid of the gross cross-section.

(6) For doubly symmetric cross-sections (e.g. $y_o = z_o = 0$), the elastic critical force $N_{cr,TF}$ for torsional-flexural buckling should be determined from:

$$N_{cr,TF} = N_{cr,T} \quad \dots (6.34)$$

provided $N_{cr,T} < N_{cr,y}$ and $N_{cr,T} < N_{cr,z}$.

(7) For cross-sections that are symmetrical about the y - y axis (e.g. $z_o = 0$), the elastic critical force $N_{cr,TF}$ for torsional-flexural buckling should be determined from:

$$N_{cr,TF} = \frac{N_{cr,y}}{2\beta} \left[1 + \frac{N_{cr,T}}{N_{cr,y}} - \sqrt{\left(1 - \frac{N_{cr,T}}{N_{cr,y}}\right)^2 + 4\left(\frac{y_o}{i_o}\right)^2 \frac{N_{cr,T}}{N_{cr,y}}} \right] \quad \dots (6.35)$$

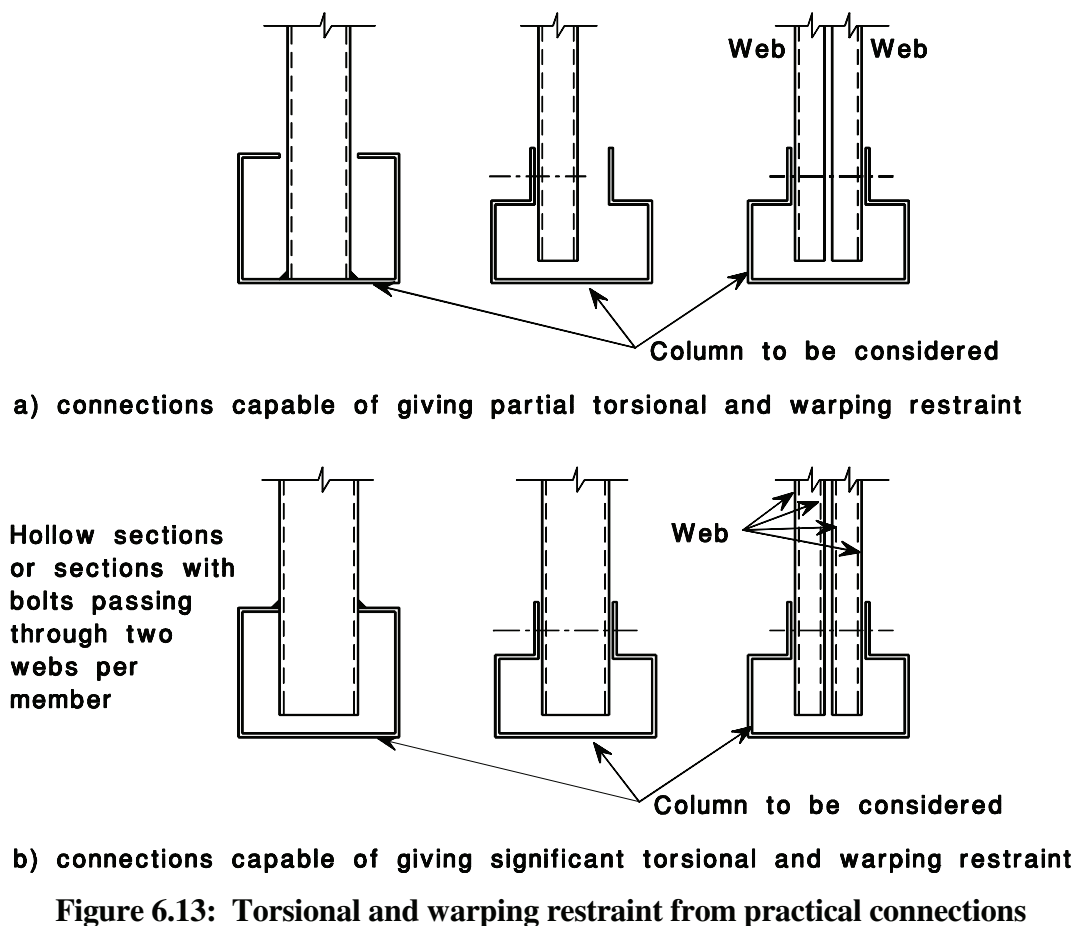
with:

$$\beta = 1 - \left(\frac{y_o}{i_o} \right)^2.$$

(8) The buckling length l_T for torsional or torsional-flexural buckling should be determined taking into account the degree of torsional and warping restraint at each end of the system length L_T .

(9) For practical connections at each end, the value of l_T/L_T may be taken as follows:

- 1,0 for connections that provide partial restraint against torsion and warping, see figure 6.13(a);
- 0,7 for connections that provide significant restraint against torsion and warping, see figure 6.13(b).



6.2.4 Lateral-torsional buckling of members subject to bending

- (1) The design buckling resistance moment of a member that is susceptible to lateral-torsional buckling should be determined according to EN 1993-1-1, section 6.3.2.2 using the lateral buckling curve b.
- (2) This method should not be used for the sections that have a significant angle between the principal axes of the effective cross-section, compared to those of the gross cross-section.

6.2.5 Bending and axial compression

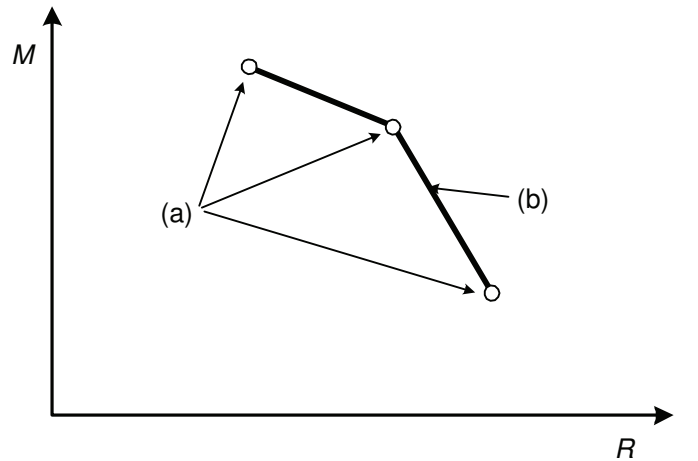
- (1) The interaction between axial force and bending moment may be obtained from a second-order analysis of the member as specified in EN 1993-1-1, based on the properties of the effective cross-section obtained from Section 5.5. See also 5.3.

(2) For each value of the test span s the support reaction R should be taken as the mean of the adjusted values of the peak load $F_{\max}$ for that value of s . The corresponding value of the support moment M should then be determined from:

$$M = \frac{s R}{4} \quad \dots (A.3)$$

Generally the influence of the dead load should be added when calculating the value of moment M following the expression (A.3).

(3) The pairs of values of M and R for each value of s should be plotted as shown in figure A.8. Pairs of values for intermediate combinations of M and R may then be determined by linear interpolation.



(a) test results for different test spans s , (b) linear interpolation

Figure A.8: Relation between support moment M and support reaction R

(4) The net deflection at the point of load application **C** in figure A.7 should be obtained from the gross measured values by deducting the mean of the corresponding deflections measured at the points **B** and **D** located at a distance e from the support points **A** and **E**, see figure A.7.

(5) For each test the applied load should be plotted against the corresponding net deflection, see figure A.9. From this plot, the rotation θ should be obtained for a range of values of the applied load using:

$$\theta = \frac{2(\delta_{pl} - \delta_e - \delta_{el})}{0,5 s - e} \quad \dots (A.4a)$$

$$\theta = \frac{2(\delta_{pl} - \delta_e - \delta_{lin})}{0,5 s - e} \quad \dots (A.4b)$$

where:

δ_{el} is the net deflection for a given load on the rising part of the curve, before $F_{\max}$;

δ_{pl} is the net deflection for the same load on the falling part of the curve, after $F_{\max}$;

δ_{lin} is the fictive net deflection for a given load, that would be obtained with a linear behaviour, see figure A.9;

δ_e is the average deflection measured at a distance e from the support, see figure A.7;

s is the test span;

e is the distance between a deflection measurement point and a support, see figure A.7.

The expression (A.4a) is used when analyses are done based on the effective cross-section. The expression (A.4b) is used when analyses are done based on the gross cross-section.

(6) The relationship between M and θ should then be plotted for each test at a given test span s corresponding to a given value of beam span L as shown in figure A.10. The design $M - \theta$ characteristic for the moment resistance of the beam over an internal support should then be taken as equal to 0,9 times the mean value of M for all the tests corresponding to that value of the beam span L .

NOTE: Smaller value than 0,9 for reduction should be used, if the full-scale tests are used to determine effect of lateral load and buckling of free flange around the mid-support, see 10.1.3.2(4).

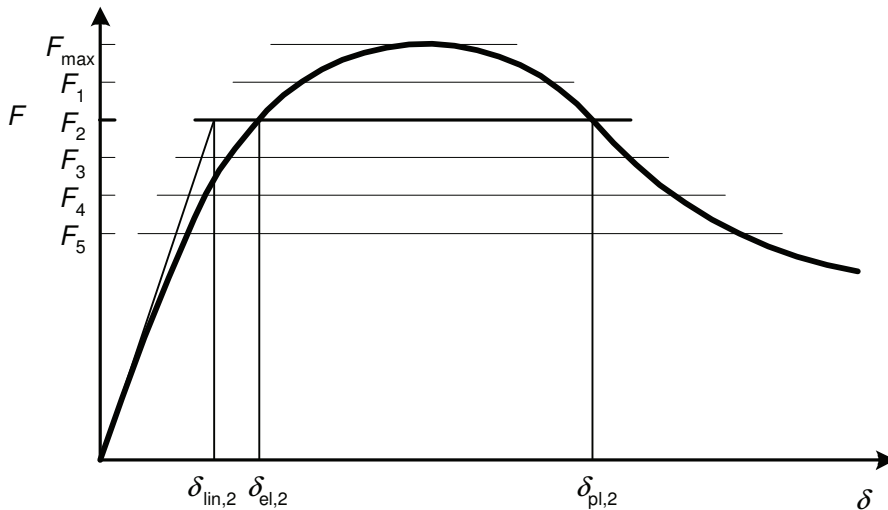
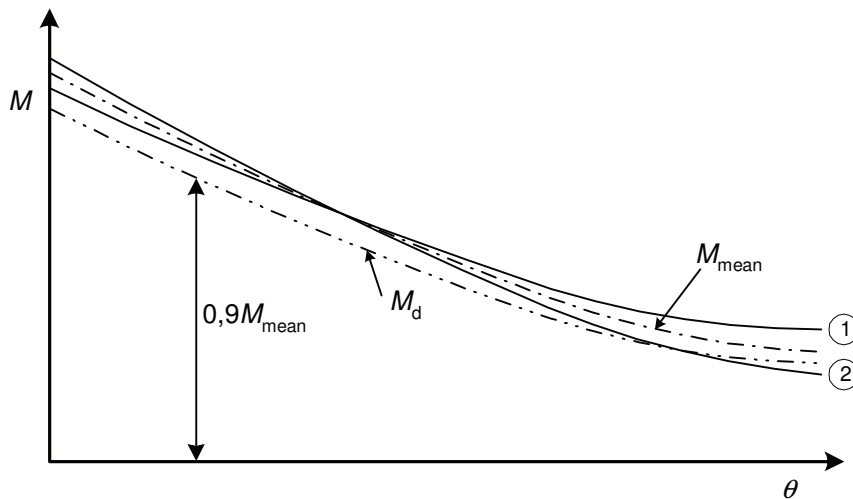


Figure A.9: Relation between load F and net deflection δ



M_{mean} = mean value, M_d = design value

Figure A.10: Derivation of moment-rotation ($M - \theta$) characteristic

A.5.3 Determination of torsional restraint

(1) The test set-up shown in figure A.11 may be used to determine the amount of torsional restraint given by adequately fastened sheeting or by another member perpendicular to the span of the beam.

(2) This test set-up covers two different contributions to the total amount of restraint as follows:

- a) The lateral stiffness K_A per unit length corresponding to the rotational stiffness of the connection between the sheeting and the beam;
- b) The lateral stiffness K_B per unit length due to distortion of the cross-section of the purlin.

(3) The combined restraint per unit length may be determined from:

$$\left(1 / K_A + 1 / K_B \right) = \delta / F \quad \dots (A.5)$$

where:

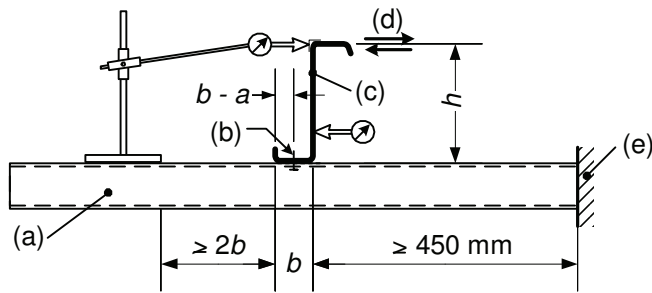
F is the load per unit length of the test specimen to produce a lateral deflection of $h/10$;

h is the overall depth of the specimen;

δ is the lateral displacement of the top flange in the direction of the load F .

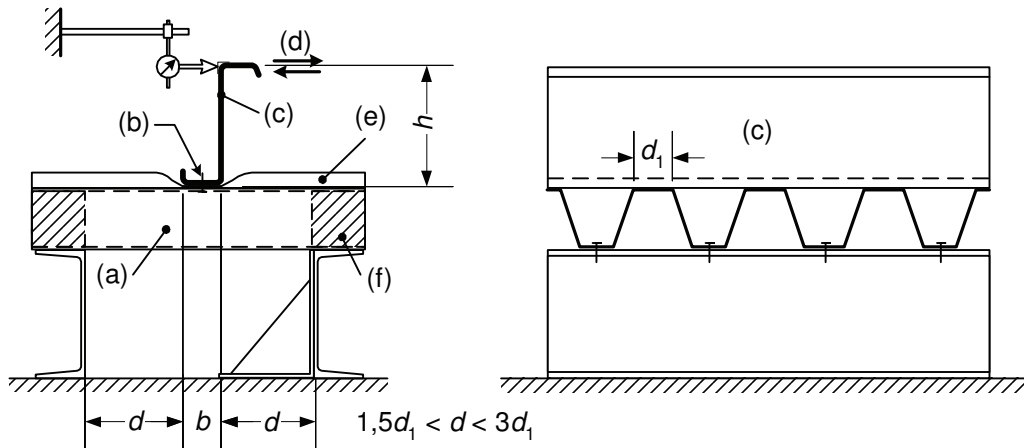
(4) In interpreting the test results, the following parameters should be treated as variables:

- the number of fasteners per unit length of the specimen;
- the type of fasteners;
- the flexural stiffness of the beam, relative to its thickness;
- the flexural stiffness of the bottom flange of the sheeting, relative to its thickness;
- the positions of the fasteners in the flange of the sheeting;
- the distance from the fasteners to the centre of rotation of the beam;
- the overall depth of the beam;
- the possible presence of insulation between the beam and the sheeting.



(a) sheeting, (b) fastener, (c) profile, (d) load, (e) clamped support

a) Alternative 1



(a) sheeting, (b) fastener, (c) profile, (d) load, (e) insulation if available, (f) timber blocks

b) Alternative 2

Figure A.11: Experimental determination of spring stiffness K_A and K_B

A.6 Evaluation of test results

A.6.1 General

(1) A specimen under test should be regarded as having failed if the applied test loads reach their maximum values, or if the gross deformations exceed specified limits.

(2) The gross deformations of members should generally satisfy:

$$\delta \leq L/50 \quad \dots (A.6)$$

$$\phi \leq 1/50 \quad \dots (A.7)$$

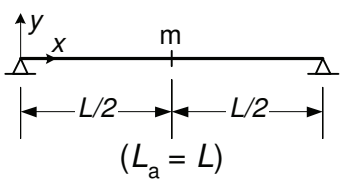
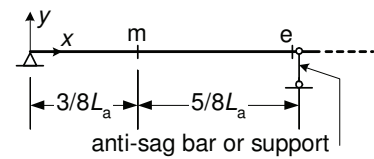
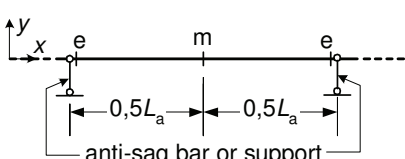
where:

δ is the maximum deflection of a beam of span L ;

ϕ is the sway angle of a structure.

(3) In the testing of connections, or of components in which the examination of large deformations is necessary for accurate assessment (for example, in evaluating the moment-rotation characteristics of sleeves), no limit need be placed on the gross deformation during the test.

Table 10.1: Values of initial moment $M_{0,fz,Ed}$ and correction factor κ_R

System	Location	$M_{0,fz,Ed}$	κ_R
	m	$\frac{1}{8} q_{h,Ed} L_a^2$	$\kappa_R = \frac{1 - 0,0225R}{1 + 0,013R}$
	m	$\frac{9}{128} q_{h,Ed} L_a^2$	$\kappa_R = \frac{1 - 0,0141R}{1 + 0,416R}$
	e	$-\frac{1}{8} q_{h,Ed} L_a^2$	$\kappa_R = \frac{1 + 0,0314R}{1 + 0,396R}$
	m	$\frac{1}{24} q_{h,Ed} L_a^2$	$\kappa_R = \frac{1 - 0,0125R}{1 + 0,198R}$
	e	$-\frac{1}{12} q_{h,Ed} L_a^2$	$\kappa_R = \frac{1 + 0,0178R}{1 + 0,191R}$

10.1.4.2 Buckling resistance of free flange

(1) If the free flange is in compression, its buckling resistance should be verified using:

$$\frac{I}{\chi_{LT}} \left(\frac{M_{y,Ed}}{W_{eff,y}} + \frac{N_{Ed}}{A_{eff}} \right) + \frac{M_{fz,Ed}}{W_{fz}} \leq f_{yb} / \gamma_{M1} \quad \dots (10.7)$$

in which χ_{LT} is the reduction factor for lateral torsional buckling (flexural buckling of the free flange).

NOTE: The use of the χ_{LT} -value may be chosen in the National Annex. The use of EN 1993-1-1, 6.3.2.3 using buckling curve b ($\alpha_{LT} = 0,34$; $\bar{\lambda}_{LT,0} = 0,4$; $\beta = 0,75$) is recommended for the relative slenderness $\bar{\lambda}_{fz}$ given in (2). In the case of an axial compression force N_{Ed} , when the reduction factor for buckling around the strong axis is smaller than the reduction factor for lateral flange buckling, e.g. in the case of many anti-sag bars, this failure mode should also be checked following clause 6.2.2 and 6.2.4.

(2) The relative slenderness $\bar{\lambda}_{fz}$ for flexural buckling of the free flange should be determined from:

$$\bar{\lambda}_{fz} = \frac{l_{fz} / i_{fz}}{\lambda_1} \quad \dots (10.8)$$

with:

$$\lambda_1 = \pi \left[E / f_{yb} \right]^{0.5}$$

where:

l_{fz} is the buckling length for the free flange from (3) to (7);

i_{fz} is the radius of gyration of the gross cross-section of the free flange plus the contributing part of the web for bending about the z-z axis, see 10.1.4.1(2).

(3) For gravity loading, provided that $0 \leq R \leq 200$, the buckling length of the free flange for a variation of the compressive stress over the length L as shown in figure 10.4 may be obtained from:

$$l_{fz} = \eta_1 L_a (1 + \eta_2 R^{\eta_3})^{\eta_4} \quad \dots (10.9)$$

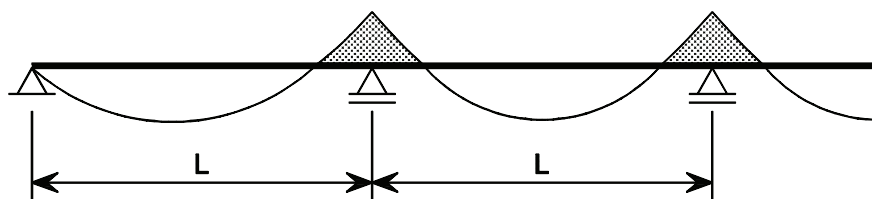
where:

L_a is the distance between anti-sag bars, or if none are present, the span L of the purlin;

R is as given in 10.1.4.1(7);

and η_1 to η_4 are coefficients that depend on the number of anti-sag bars, as given in table 10.2a. The tables 10.2a and 10.2b are valid only for equal spans uniformly loaded beam systems without overlap or sleeve and with anti-sag bars that provide lateral rigid support for the free flange. The tables may be used for systems with sleeves and overlaps provided that the connection system may be considered as fully continuous. In other cases the buckling length should be determined by more appropriate calculations or, except cantilevers, the values of the table 10.2a for the case of 3 anti-sag bars per field may be used.

NOTE: Due to rotations in overlap or sleeve connection, the field moment may be much larger than without rotation which results also in longer buckling lengths in span. Neglecting the real moment distribution may lead to unsafe design.



[Dotted areas show regions in compression]

Figure 10.4: Varying compressive stress in free flange for gravity load cases

(5) The lateral bending moment $M_{fz,Ed}$ may be determined from expression (10.5) except for a beam with the free flange in tension, where, due to positive influence of flange curling and second order effect moment $M_{fz,Ed}$ may be taken equal to zero:

$$M_{fz,Ed} = \kappa_R M_{0,fz,Ed} \quad \dots (10.5)$$

where:

$M_{0,fz,Ed}$ is the initial lateral bending moment in the free flange without any spring support;

κ_R is a correction factor for the effective spring support.

(6) The initial lateral bending moment in the free flange $M_{0,fz,Ed}$ may be determined from table 10.1 for the critical locations in the span, at supports, at anti-sag bars and between anti-sag bars. The validity of the table 10.1 is limited to the range $R \leq 40$.

(7) The correction factor κ_R for the relevant location and boundary conditions, may be determined from table 10.1 (or using the theory of beams on the elastic Winkler foundation), using the value of the coefficient R of the spring support given by:

$$R = \frac{K L_a^4}{\pi^4 E I_{fz}} \quad \dots (10.6)$$

where:

I_{fz} is the second moment of area of the gross cross-section of the free flange plus the contributing part of the web for bending about the z-z axis, see 10.1.4.1(2); when numerical analysis is carried out, see 10.1.2(5);

K is the lateral spring stiffness per unit length from 10.1.5.1;

L_a is the distance between anti-sag bars, or if none are present, the span L of the purlin.

Table 10.2a : Coefficients η_i for down load with 0, 1, 2, 3, 4 anti-sag bars

Situation	Anti sag-bar Number	η_1	η_2	η_3	η_4
End span	0	0.414	1.72	1.11	-0.178
Intermediate span		0.657	8.17	2.22	-0.107
End span	1	0.515	1.26	0.868	-0.242
Intermediate span		0.596	2.33	1.15	-0.192
End and intermediate span	2	0.596	2.33	1.15	-0.192
End and intermediate span	3 and 4	0.694	5.45	1.27	-0.168

Table 10.2b : Coefficients η_i for uplift load with 0, 1, 2, 3, 4 anti-sag bars

Situation	Anti sag-bar Number	η_1	η_2	η_3	η_4
Simple span	0	0.694	5.45	1.27	-0.168
End span		0.515	1.26	0.868	-0.242
Intermediate span		0.306	0.232	0.742	-0.279
Simple and end spans	1	0.800	6.75	1.49	-0.155
Intermediate span		0.515	1.26	0.868	-0.242
Simple span	2	0.902	8.55	2.18	-0.111
End and intermediate spans		0.800	6.75	1.49	-0.155
Simple and end spans	3 and 4	0.902	8.55	2.18	-0.111
Intermediate span		0.800	6.75	1.49	-0.155

(4) For gravity loading, if there are more than three equally spaced anti-sag bars, and under conditions specified in (3), the buckling length need not be taken as greater than the value for two anti-sag bars, with $L_a = L/3$. This clause is valid only if there is no axial compressive force.

(5) If the compressive stress over the length L is almost constant, due to the application of a relatively large axial force, the buckling length should be determined using the values of η_i from table 10.2a for the case shown as more than three anti-sag bars per span, but the actual spacing L_a .

(6) For uplift loading, when anti-sag bars are not used, provided that $0 \leq R_0 \leq 200$, the buckling length of the free flange for variations of the compressive stress over the length L_0 as shown in figure 10.5, may be obtained from:

$$l_{fz} = 0,7 L_0 (1 + 13,1 R_0^{1,6})^{-0,125} \quad \dots (10.10a)$$

with:

$$R_0 = \frac{K L_0^4}{\pi^4 E I_{fz}} \quad \dots (10.10b)$$

in which I_{fz} and K are as defined in 10.1.4.1(7). Alternatively, the buckling length of the free flange may be determined using the table 10.2b in combination with the equation given in 10.1.4.2(3).

(7) For uplift loading, if the free flange is effectively held in position laterally at intervals by anti-sag bars, the buckling length may conservatively be taken as that for a uniform moment, determined as in (5). The formula (10.10a) may be applied under conditions specified in (3). If there are no appropriate calculations, reference should be made to 10.1.4.2(5).

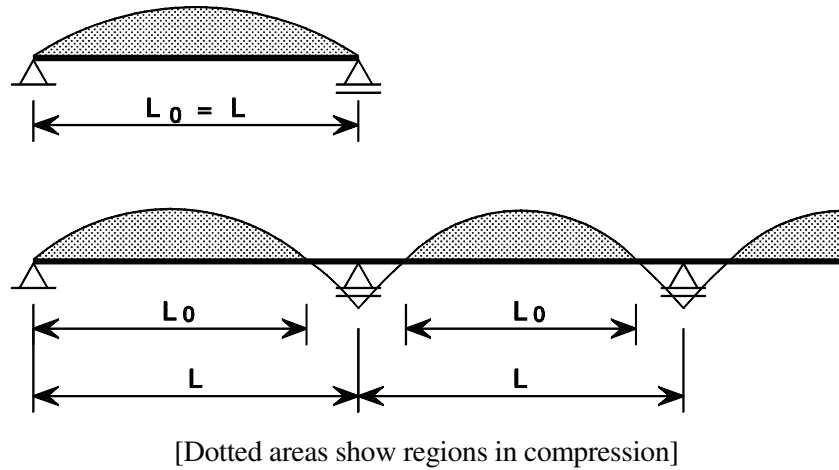


Figure 10.5: Varying compressive stress in free flange for uplift cases

10.1.5 Rotational restraint given by the sheeting

10.1.5.1 Lateral spring stiffness

(1) The lateral spring support given to the free flange of the purlin by the sheeting should be modelled as a lateral spring acting at the free flange, see figure 10.1. The total lateral spring stiffness K per unit length should be determined from:

$$\frac{1}{K} = \frac{1}{K_A} + \frac{1}{K_B} + \frac{1}{K_C} \quad \dots (10.11)$$

where:

K_A is the lateral stiffness corresponding to the rotational stiffness of the joint between the sheeting and the purlin;

K_B is the lateral stiffness due to distortion of the cross-section of the purlin;

K_C is the lateral stiffness due to the flexural stiffness of the sheeting.

(2) Normally it may be assumed to be safe as well as acceptable to neglect $1/K_C$ because K_C is very large compared to K_A and K_B . The value of K should then be obtained from:

$$K = \frac{1}{(1/K_A + 1/K_B)} \quad \dots (10.12)$$

(3) The value of $(1/K_A + 1/K_B)$ may be obtained either by testing or by calculation.

NOTE: Appropriate testing procedures are given in Annex A.

(2) The design value of the bending moment M_{Ed} should satisfy

$$\frac{M_{Ed}}{M_{LT,Rd}} \leq 1 \quad \dots(E.1)$$

where

$$M_{LT,Rd} = \left(\frac{f_y}{\gamma_{M1}} \right) W_{eff,y} \frac{\chi_{LT}}{k_d} \quad \dots(E.2)$$

and

$W_{eff,y}$ is section modulus of the effective cross-section with regard to the y-y axis;

χ_{LT} is reduction factor for lateral torsional buckling in dependency of $\bar{\lambda}_{LT}$ due to 6.2.3, where α_{LT} is substituted by $\alpha_{LT,eff}$;

and

$$\bar{\lambda}_{LT} = \sqrt{\frac{W_{eff,y} f_y}{M_{cr}}} \quad \dots(E.3)$$

$$\alpha_{LT,eff} = \alpha_{LT} \sqrt{\frac{W_{el,y}}{W_{eff,y}}} \quad \dots(E.4)$$

and

α_{LT} is imperfection factor due to 6.2.3;

$W_{el,y}$ is section modulus of the gross cross-section with regard to the y-y axis;

k_d is coefficient for consideration of the non restraint part of the purlin according to equation (E.5) and table E.2;

$$k_d = \left(a_1 - a_2 \frac{L}{h} \right), \text{ but } \geq 1.0 \quad \dots(E.5)$$

a_1, a_2 coefficients from table E.2;

L span of the purlin;

h overall depth of the purlin.

Table E.2: Coefficients a_1, a_2 for equation (E.5)

System	Z-purlins		C-purlins		Σ -purlins	
	a_1	a_2	a_1	a_2	a_1	a_2
single span beam gravity load	1.0	0	1.1	0.002	1.1	0.002
single span beam uplift load	1.3	0	3.5	0.050	1.9	0.020
continuous beam gravity load	1.0	0	1.6	0.020	1.6	0.020
continuous beam uplift load	1.4	0.010	2.7	0.040	1.0	0

(3) The reduction factor χ_{LT} may be chosen by equation (E.6), if a single span beam under gravity load is present or if equation (E.7) is met

$$\chi_{LT} = 1,0 \quad \dots(E.6)$$

$$C_D \geq \frac{M_{el,u}^2}{E I_v} k_{\vartheta} \quad \dots(E.7)$$

where

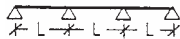
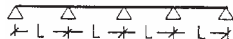
$$M_{el,u} = W_{el,u} f_y \quad \text{elastic moment of the gross cross-section with regard to the major u-u axis; } \dots(E.8)$$

$$I_v \quad \text{moment of inertia of the gross cross-section with regard to the minor v-v axis:}$$

$$k_{\vartheta} \quad \text{factor for considering the static system of the purlin due to table E.3.}$$

NOTE: For equal flanged C-purlins and Σ -purlins $I_v = I_z$, $W_u = W_y$, and $M_{el,u} = M_{el,y}$. Conventions used for cross section axes are shown in Figure 1.7 and section 1.6.4.

Table E.3: Factors k_{ϑ}

Statcal system	Gravity load	Uplift load
	-	0.210
	0.07	0.029
	0.15	0.066
	0.10	0.053

(4) The reduction factor χ_{LT} should be calculated according to 6.2.4 using $\bar{\lambda}_{LT}$ and $\alpha_{LT,eff}$ in cases which are not met by (3). The elastic critical moment for lateral-torsional buckling M_{cr} may be calculated by the equation (E.9):

$$M_{cr} = \frac{k}{L} \sqrt{G I_t^* E I_v} \quad \dots(E.9)$$

where

I_t^* is the fictitious St. Venant torsion constant considering the effective rotational restraint by equation (E.10) and (E.11):

$$I_t^* = I_t + C_D \frac{L^2}{\pi^2 G} \quad \dots(E.10)$$

I_t is St. Venant torsion constant of the purlin;

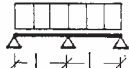
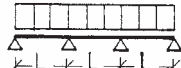
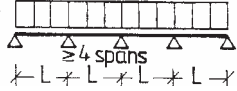
$$1/C_D = \frac{1}{C_{D,A}} + \frac{1}{C_{D,B}} + \frac{1}{C_{D,C}} \quad \dots(E.11)$$

$C_{D,A}$, $C_{D,C}$ rotational stiffnesses due to 10.1.5.2;

$C_{D,B}$ rotational stiffnesses due to distortion of the cross-section of the purlin due to 10.1.5.1, $C_{D,B} = K_B h^2$, where h = depth of the purlin and K_B according to 10.1.5.1;

k lateral torsional buckling coefficient due to table E.4.

Table E.4: Lateral torsional buckling coefficients k for purlins restrained horizontally at the upper flange

Statical system	Gravity load	Uplift load
	∞	10.3
	17.7	27.7
	12.2	18.3
	14.6	20.5